

Chapter 3

Network Fulfilment Models

3.1 Introduction

So far, we have described mathematical models to match the random demand in the time dimension. In this Chapter, we change the pace by thinking of problems satisfying the demand physically in the space dimension. Most if not all problems in supply chain analysis deal with some form of transportation problem as the demand requires moving goods or supplies between the producers and customers.

Matching the demand in space also adds a non-trivial time delay that needs to be accounted for when matching resources to satisfy the demand. Large companies like Amazon have to optimize the location of their warehouses and orchestrate a large chain of distribution networks to deliver their products in record time. Likewise, ridesharing marketplaces (such as Uber or Doordash) have to balance a demand that changes in space and time with a supply side of workers and drivers that will service different geographical locations as they finish their deliveries and rides.

We will explore a series of common problems and solutions to supply chain management problems in space, as well as algorithms and heuristics to solve them.

3.2 Warehouse consolidation and positioning

Imagine a supplier wanting to build a warehouse to service demand in a city or geographical location. The supplier would want to build the warehouse in a place that is economically viable and resource efficient to service this demand.

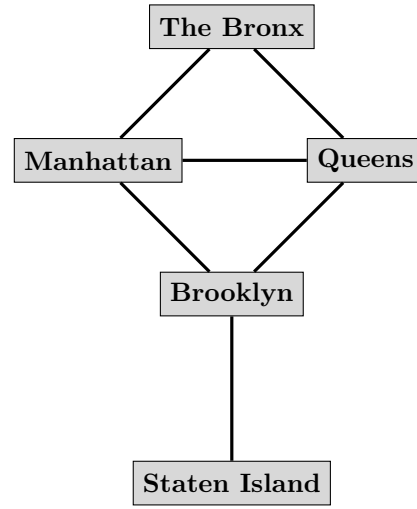
While the physical world that we inhabit is 3-dimensional, it is often useful to think of abstractions that simplify the problem: For example, most geographical maps use a 2-dimensional

representation of the Earth (this makes sense as the Earth is more or less a manifold locally, i.e. locally, we live in a flat world). Perhaps, one of the most powerful abstractions for physical spaces are graphs. A graph G is defined as a set of vertices V and edges E .

Graphs can be seen as discretization of a continuous space into a disjoint partition, connecting with edges all physically adjacent partitions. For example, take a map such as in the left side of Figure 3.1 with the five Boroughs of New York City (Manhattan, The Bronx, Queens, Brooklyn and Staten Island) and make each Borough a vertex, drawing an edge if there is a road connection between two Boroughs (vertices).



(a) Geographic Map of NYC's Five Boroughs.



(b) Graph abstraction of NYC.

Figure 3.1: Graph abstraction of NYC representing each of the five Boroughs as vertices, and their roads connections as edges.

Modeling a physical space as a graph has the natural trade-off of having tractable algorithms to solve a wide-range of problems while losing some of the accuracy and resolution of the underlying modeled real-world system. For example, in the graph of the figure, direct travels between Manhattan and Queens can occur via two different bridges, but there is only one edge representing their connection. In modeling traveling times, there is either the choice of creating more nodes and edges to represent the different routes and to attach numerically the respective travel time to each edge, or to *average* into the single edge all the possible routes between Queens and Manhattan. Clearly, the tradeoff is between accuracy of the abstraction and tractability of the solution, as most optimization algorithms scale proportional to the size of the Graph. In general, mathematical modeling is the art of choosing an appropriate abstraction of a real-life system that preserves the behavior of interest while being algorithmically tractable.

Going back to the problem of choosing a location for a warehouse, start with a graph G , composed by N vertices $V = \{1, 2, \dots, N\}$ and edges $E \subseteq \{(i, j) | (i, j) \in V^2\}$. For example, the graph in Figure 3.1 is composed by $N = 5$ vertices $V = \{1, 2, 3, 4, 5\}$, each vertex corresponds

to a Borough: 1 $\rightarrow$ Manhattan, 2 $\rightarrow$ The Bronx, 3 $\rightarrow$ Queens, 4 $\rightarrow$ Brooklyn and 5 $\rightarrow$ Staten Island. And edges $E = \{(1, 2), (1, 3), (1, 4), (2, 3), (3, 4), (4, 5)\}$.

Vertices can have different unique attributes, for example, in the model of this section each vertex has an associated random demand D_i for $i = 1, \dots, N$. This demand, is the usual demand random variable and denotes the random realization of the demand (in this case, there is not yet a temporal element to the demand and it can be taken as a static model with i.i.d. realizations of the demand that repeat every day). Likewise, every edge (i, j) for $i, j \in \{1, \dots, N\}$ also has a random variable T_{ij} denoting the random traveling time of a vehicle starting in vertex i to vertex j .

3.2.1 Single Location Warehouse

With these elements, we can now start to model the problem of choosing a location to build a warehouse. Let c_i be the cost of running a warehouse in location i with enough capacity to service the demand $\sum_{i=1}^N D_i$ with high probability (see Chapters 1 and 2 on how this number can be calculated, here it is taken as given¹). In this scenario, if it's not possible to run a warehouse in some vertex, then let the cost be $c_i = \infty$ for such vertex (or exclude it from the set of possible locations to run the warehouse). Likewise, let the cost of transports between vertices (i, j) be equal to² $c_{ij} \propto T_{ij}$, for example the cost of the labor of a courier plus the cost of gas that is proportional to the random traveling time T_{ij} , that is $c_{ij} = \gamma_{ij}T_{ij}$, clearly c_{ij} is also a random variable.

For a warehouse built on location i , the cost is equal to $c_i + \sum_{j=1}^N D_j c_{ij}$. As this number is a random variable, a first approximation is to look at the expected value and choose over all possible locations, that is, solving the enumeration problem:

$$\min_{i=1, \dots, N} \mathbb{E} \left[c_i + \sum_{j=1}^N D_j c_{ij} \right], \quad (3.1)$$

It is an enumeration problem because it amounts to calculate the cost for each location and choose the minimum. The emphasis in writing the costs as random variables instead of the more common practice of writing them as constants is that there is a conscious choice to solve the problem only in expected value or to consider other moments of their distribution. For example, it could be the case that running the warehouse in one location has the lowest cost in expected value, but the variation (either of the demands and the trips) is higher than another location that has a slightly higher cost in expected value, but less variance (in this case, variance denotes

¹In practice, one way would be to calculate a capacity ϵ such that $\mathbb{P}(\sum_{i=1}^N D_i > \epsilon) \leq 1 - \alpha$ and then quote or calculate the actual cost of running a warehouse of that capacity for the value of c_i . You can also set up a Newsvendor-like model to calculate a capacity that is also sensitive to missed sales or holding costs.

² $a \propto b$ denotes that b is proportional to a up to a constant, say $b = 2a$.

the daily variation of running the warehouse satisfying random demands every day or period). See Figure 3.2 to get a feel for evaluating the cost of building a single warehouse in Manhattan.

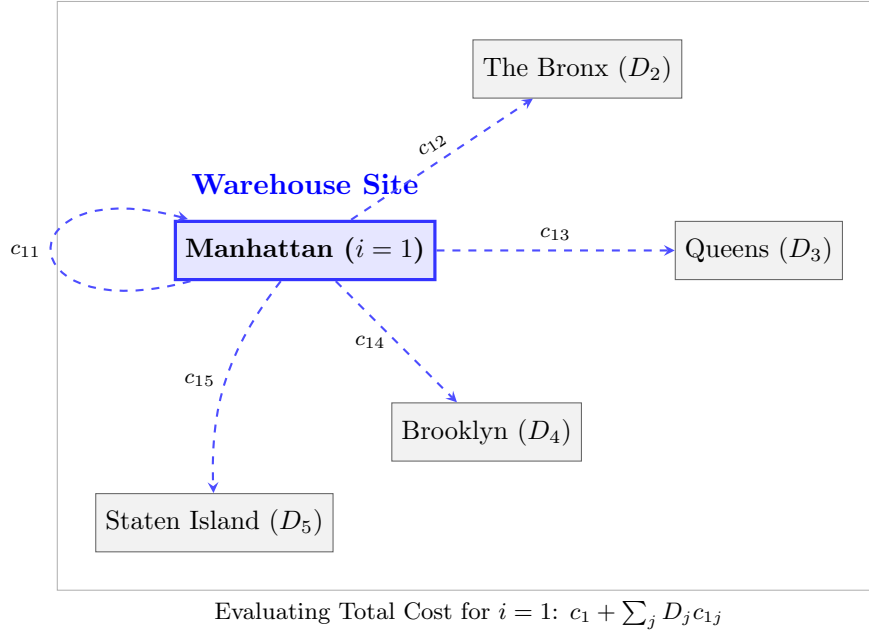


Figure 3.2: Visualization of the enumeration step for $i = 1$ (Manhattan). To solve the problem, the supplier calculates the expected total cost (warehouse overhead c_i plus weighted transportation costs to all demand nodes $D_j c_{ij}$) for each potential vertex and selects the minimum.

In general, for problems involving a random amount of money (either a cost or a payoff) there is an amount of risk aversion that the decision-maker is willing to take, and it needs to be made explicit in the problem formulation, or otherwise be acknowledged by the modeler. For example, given two lotteries: one that pays $X_1 = \$100,000B_1 + (-\$100,000)(1 - B_1)$ and another $X_2 = \$100B_2 + (-\$100)(1 - B_2)$ with two independent coins B_1, B_2 with $P(B = 1) = P(B = 0) = 1/2$. That is, two lotteries that with 50% chance pay a positive amount and with the same chance the player loses the same amount, in the first case \$100,000 and in the other \$100. While both lotteries pay in average \$0, that is $E(X_1) = E(X_2) = \$0$ they are clearly not the same game. Depending on their risk aversion, some players would choose to play lottery 1, while others would play lottery 2 (or even choose not to play at all). The quantitative difference between the two games is their variance. We have that the variance of the first lottery is $\text{Var}(X_1) = 2^2(\$100,000)^2\text{Var}(B_1) = (\$100,000)^2$ and the second one $\text{Var}(X_2) = (\$100)^2$. The risk ratio between the two lotteries is $\text{Var}(X_1)/\text{Var}(X_2) = 10^6$. In other words, playing the first lottery has a million times higher variance than the second. While exaggerated, this example shows the potential literal risk of only modeling decisions based on their expected value. A quantitative approach to evaluate each lottery could be to *weight* the variance or the standard deviation, for example $E(X_1) + \lambda\sqrt{\text{Var}(X_1)}$ weights the standard deviation (in dollars). A weight $\lambda = 0$ denotes risk indifference (thus, only caring for the expected return of

the lottery). $\lambda < 0$ denotes risk aversion, while $\lambda > 0$ implies risk appetite. Note that the units of $E(X_1) + \lambda\sqrt{\text{Var}(X_1)}$ is dollars.

Another justification coming from the study of risk measures also justifies this by defining the risk measure of a random variable X as $R_\lambda(X) = \frac{1}{\lambda} \ln E(e^{\lambda X})$. This risk measure is used in risk management as it captures risk aversion, is convex and coherent and it is sensitive to tail risks. In fact, we show that our strategy of weighting the risk is approximately equivalent to this risk measure. To see why, note that $e^u = 1 + u + u^2/2 + \dots$. Then, $E(e^{\lambda X}) = E(1 + \lambda X + \frac{\lambda^2 X^2}{2} + \dots) = 1 + \lambda E(X) + \frac{\lambda^2}{2} E(X^2) + \dots = 1 + v$ where $v = \lambda E(X) + \frac{\lambda^2}{2} E(X^2) + \dots$, also note that $\ln(1 + v) = v - \frac{v^2}{2} + \dots$. Using these relations and noting that terms with v^3 and higher vanish, we get:

$$\begin{aligned} \ln E(e^{\lambda X}) &= \lambda E(X) + \frac{\lambda^2}{2} E(X^2) - \frac{1}{2} \left[\lambda E(X) + \frac{\lambda^2}{2} E(X^2) \right]^2 + \dots \\ &= \lambda E(X) + \frac{\lambda^2}{2} E(X^2) - \frac{\lambda^2}{2} [E(X)]^2 + \dots \\ &= \lambda E(X) + \frac{\lambda^2}{2} [E(X^2) - E(X)^2] + \dots \\ &= \lambda E(X) + \frac{\lambda^2}{2} \text{Var}(X) + O(\lambda^3) \end{aligned} \quad (3.2)$$

Noting that the term $O(\lambda^3)$ vanishes to 0 and dividing both sides by λ yields the desired relation.

This shows that optimizing the objective $E(X) + \lambda \text{Var}(X)$ is a principled way of handling the risk. An important aspect in practice is calibrating the risk aversion parameter λ . A well-known principle to do this, is known as the *Certainty Equivalent*. This is simply an amount of money that the decision maker is willing to pay (or receive) to be indifferent between the lottery and a certain payoff. Solving for that amount would yield the implied risk aversion coefficient. For example, suppose there is a lottery $X = -\$100B + \$0(1 - B) = -\$100B$, that is, losing a \$100 with probability $P(B = 1)$ or losing 0 with probability $P(B = 0)$. Suppose $P(B = 1) = 1/2$, we have $E(X) = -\$50$ and $\text{Var}(X) = (\$100)^2/4$. A person is willing to pay \$40 to exit the lottery, this implies a Certainty Equivalent (CE) given by $-40 = -50 + \frac{\lambda}{8} 100^2$ or $\lambda = 80/100^2$. In general, giving the stakeholders similar lotteries and with their CEs is a standard way to learn the risk aversion of a stakeholder, we will illustrate numerically how the trade-off between risk and expected cost plays out later.

Going back to the warehouse location problem, the problem incorporating variance can be stated as follows:

$$\min_{i=1,\dots,N} E \left[c_i + \sum_{j=1}^N D_j c_{ij} \right] + \lambda \sqrt{\text{Var} \left(\sum_{j=1}^N D_j c_{ij} \right)}, \quad (3.3)$$

With $\lambda > 0$ this optimization problem models a risk-averse decision-maker that penalizes variance in their cost (caused by both variance in the demand and in the transportation times).

This is again an enumeration problem that amounts to calculate the objective for each potential warehouse location. It is also possible to consider other risk measures that penalize variation.

In the case where the cost of the facilities c_i is fixed. The variance of the cost $\text{Var}[c_k + \sum_{j=1}^N D_j c_{kj}]$ can be calculated as $\text{Var}[\sum_{j=1}^N D_j c_{kj}] = \mathbf{1} \Sigma_k \mathbf{1}^\top$ where the entry i, j of the matrix Σ_k is $\text{Cov}(D_i c_{ki}, D_j c_{kj}) = \gamma_{ki} \gamma_{kj} \text{Cov}(D_i T_{ki}, D_j T_{kj})$. For example, in the case of the diagonal we have $\text{Cov}(D_i c_{ki}, D_i c_{ki}) = \gamma_{ki}^2 \text{Var}(D_i T_{ki})$ if the demand and the travel times are independent then this is equal to $\gamma_{ki}^2 \text{Var}(D_i) \text{Var}(T_{ki})$. For the other elements off-diagonal we use the definition of covariance³, then, $\text{Cov}(D_i T_{ki}, D_j T_{kj}) = \text{E}(D_i T_{ki} D_j T_{kj}) - \text{E}(D_i T_{ki}) \text{E}(D_j T_{kj})$. With the assumption that while the demands and the travel times are independent, but not the demands and the travel times with each other. For example, it makes sense that travel jams in Manhattan could affect traffic in The Bronx, and also that the demand for a product in Brooklyn is correlated with the demand for the same product in Queens. With this assumption, we have that $\text{E}(D_i T_{ki} D_j T_{kj})$ is equal to $\text{E}(D_i D_j) \text{E}(T_{ki} T_{kj})$ and $\text{E}(D_i T_{ki}) \text{E}(D_j T_{kj})$ is equal to $\text{E}(D_i) \text{E}(T_{ki}) \text{E}(D_j) \text{E}(T_{kj})$. In summary, the covariance matrix Σ_k has entries i, j equal to:

$$\Sigma_k(i, j) = \begin{cases} \gamma_{ki}^2 \text{Var}(D_i) \text{Var}(T_{ki}), & \text{for } i = j \\ \gamma_{ki} \gamma_{kj} [\text{E}(D_i D_j) \text{E}(T_{ki} T_{kj}) - \text{E}(D_i) \text{E}(D_j) \text{E}(T_{ki}) \text{E}(T_{kj})], & \text{for } i \neq j. \end{cases} \quad (3.4)$$

3.2.2 Multiple Location Warehouses

Sometimes it can be more efficient to choose multiple locations to satisfy the geographical demand of multiple locations than a single warehouse location. This can be because the cost structure is more favorable (closer deliveries are cheaper), due to economies of scale, or simply ease of logistical burden by breaking down the demand to multiple different warehouses. The problem can also be interpreted as choosing providers to satisfy demand for different geographical locations.

To modify the previous section formulation, now each candidate location has a finite capacity C_i to service demand. Likewise, we modify the cost c_i to be a fixed (and given) cost of running the warehouse i . The second and more important modification is that since multiple locations can be chosen to service the demand, brute enumeration is in general not a great strategy, except for small instances of the problem. To see why, suppose there are M candidate locations, all possible choices of subsets of locations is 2^M , which is a number that grows exponentially in the number of possible locations⁴. Combinatorial Optimization studies algorithms to solve problems where the solution takes the form of subsets by exploiting the internal structure of the problem, not to be forced to enumerate all possible 2^M candidate solutions. A very powerful technique to solve this problems is called Mixed Integer Programming, that is very similar to Linear Programming (both in structure and algorithmically) by allowing integer variables (normally

³ $\text{Cov}(X, Y) = \text{E}(XY) - \text{E}(X)\text{E}(Y)$.

⁴Given a set of M elements, each can be either included or not included in a subset. Since this is true for all elements, the total number of subsets is 2^M , also called the *power set*.

encoding either counting quantities or coding yes-no decisions as variables). See Figure 3.3 for a typical representation of the problem of matching possible M warehouses locations with N demand target locations.

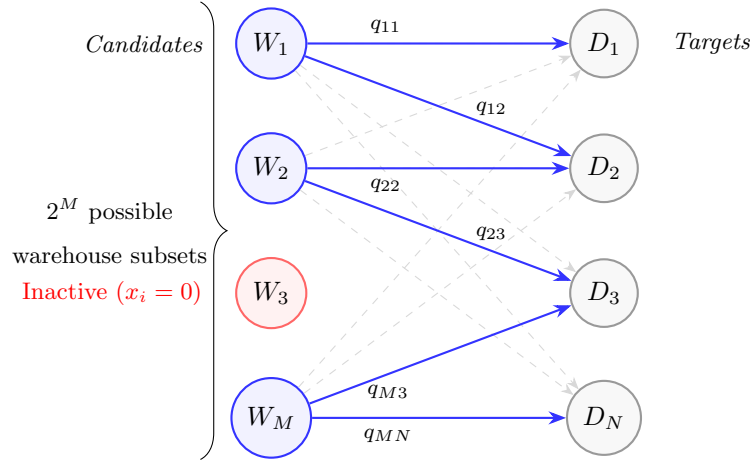


Figure 3.3: Bipartite representation of the Multiple Location Warehouse problem. Solid blue lines represent an active assignment q_{ij} from selected warehouses ($x_i = 1$). The light dashed lines represent the combinatorial complexity: the model must select a subset of M locations and then optimize the flows to N targets.

A Mixed Integer Programming formulation of this problem entails adding binary variables $x_i \in \{0, 1\}$ denoting whether or not the location is chosen. Also, the variables $q_{ij} \geq 0$ denote the amount of demand (quantity) satisfied from warehouse i to location $j = 1, \dots, N$. The next modeling step is to relate variables x_i and q_{ij} to each other in an equation (that normally becomes a constraint of the problem). For each warehouse it must be true that when it is selected ($x_i = 1$), the total quantity that can be serviced from the location cannot exceed its capacity C_i . In other words, $\sum_{j=1}^N q_{ij} \leq C_i$ when $x_i = 1$, and conversely when $x_i = 0$ it should be $\sum_{j=1}^N q_{ij} = 0$. MIP formulations normally involve tricks manipulating real-world conditions into equations (see [1] for some of the most common tricks). In this case, the trick is to write the constraint as $\sum_{j=1}^N q_{ij} \leq x_i C_i$. It is immediate that this constraint achieves the goal of encoding both situations.

Likewise, it must be true that the random demand is satisfied to some level. For example, if looking at the problem in expected value, that the demand is satisfied in average is the constraint $\sum_{i=1}^M q_{ij} \geq E(D_j)$. Another common technique to deal with this type of constraints is to write a constraint as $P(\sum_{i=1}^M q_{ij} \leq D_j) \leq \eta$, meaning that with probability not exceeding η (say a small percentage like 5%) the random demand will be satisfied. In the case where the demand can be approximated as a normal random variable, that is, $D_j \stackrel{d}{=} E(D_j) + \sqrt{\text{Var}(D_j)}Z$. The constraint $P(\sum_{i=1}^M q_{ij} \leq D_j) \leq \eta$ can then be written as $P(Z \leq [\sum_{i=1}^M q_{ij} - E(D_j)] / \sqrt{\text{Var}(D_j)}) \geq 1 - \eta$, or the constraint $\sum_{i=1}^M q_{ij} \geq E(D_j) + \sqrt{\text{Var}(D_j)}\Phi^{-1}(1 - \eta)$, the similarity to the Newsvendor model studied in Chapter 2 should be evident, as the constraint conveys that the quantity delivered

should be the mean of the demand plus a buffer proportional to the standard deviation of the demand. The last constraint to be enforced is that $q_{ij} = 0$ when $x_i = 0$ and positive otherwise. Letting $\bar{d}_j$ be an upper bound for the demand such that $P(D_j > \bar{d}_j) = 0$, this can be achieved with the constraint $q_{ij} \leq x_i \bar{d}_j$. With these constraints, we can write a first fomulation of the problem in expected value as (let $W \subseteq V$ be the set of possible locations for the warehouses to make the notation cleaner):

$$\begin{aligned} \min_{q_{ij} \geq 0, x_i \in \{0,1\}} \quad & \mathbb{E} \left\{ \sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right\} \\ \text{s.t.} \quad & \sum_{i \in W} q_{ij} \geq \mathbb{E}(D_j) \text{ for } j \in V, \\ & \sum_{j \in V} q_{ij} \leq x_i C_i \text{ for } i \in W, \\ & q_{ij} \leq x_i \bar{d}_j \text{ for } i, j \in W \times V. \end{aligned} \quad (3.5)$$

As the costs c_i are fixed, the objective can be written as $\sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} \mathbb{E}(c_{ij}) q_{ij}$. This formulation is the typical warehouse optimization found in textbooks and can be solved with off-the-shelf MIP solvers.

Similar to the discussion of the single warehouse model, as is, the model is only minimizing the cost in expected value (both in the objective and the first constraint). Variations in the travel times T_{ij} will cause fluctuations in the cost. We can take a similar strategy to hedge this variation using a similar reasoning as in the single warehouse case. To do so, note that the variance $\text{Var}[\sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij}]$ is equal to $\sum_{i \in W} \sum_{j \in V} \sum_{k \in W} \sum_{l \in V} q_{ij} \text{Cov}(c_{ij}, c_{lk}) q_{lk}$ which can be written in matrix form as $\mathbf{q} \Sigma \mathbf{q}^T$ where Σ has entries:

$$\text{Cov}(c_{ij}, c_{lk}) = \begin{cases} \gamma_{ij}^2 \text{Var}(T_{ij}), & \text{for } i, j = k, l \\ \gamma_{ij} \gamma_{kl} [\mathbb{E}(T_{ij} T_{kl}) - \mathbb{E}(T_{ij}) \mathbb{E}(T_{kl})], & \text{for } i, j \neq k, l. \end{cases} \quad (3.6)$$

Also using the chance constraint for the demand, the risk aware formulation of the problem can be written as a Mixed Integer Quadratic Program (MIQP):

$$\begin{aligned} \min_{q_{ij} \geq 0, x_i \in \{0,1\}} \quad & \mathbb{E} \left\{ \sum_{i \in W} c_i x_i + \sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right\} + \lambda \text{Var} \left(\sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij} \right) \\ \text{s.t.} \quad & \sum_{i \in W} q_{ij} \geq \mathbb{E}(D_j) + \sqrt{\text{Var}(D_j)} \Phi^{-1}(1 - \eta) \text{ for } j \in V, \\ & \sum_{j \in V} q_{ij} \leq x_i C_i \text{ for } i \in W, \\ & q_{ij} \leq x_i \bar{d}_j \text{ for } i, j \in W \times V. \end{aligned} \quad (3.7)$$

This formulation of the problem can be solved with a MIQP solver. Note that in this case the objective uses the variance instead of the standard deviation as in the single warehouse section. The problem can still be solved as a Mixed Integer Second-Order Conic Program (MISOCP). The formulation is also implicitly assuming that the travel times and the demands

are independent from each other. In Figure 3.4 we can see the trade-off we mentioned earlier considering different values of λ .

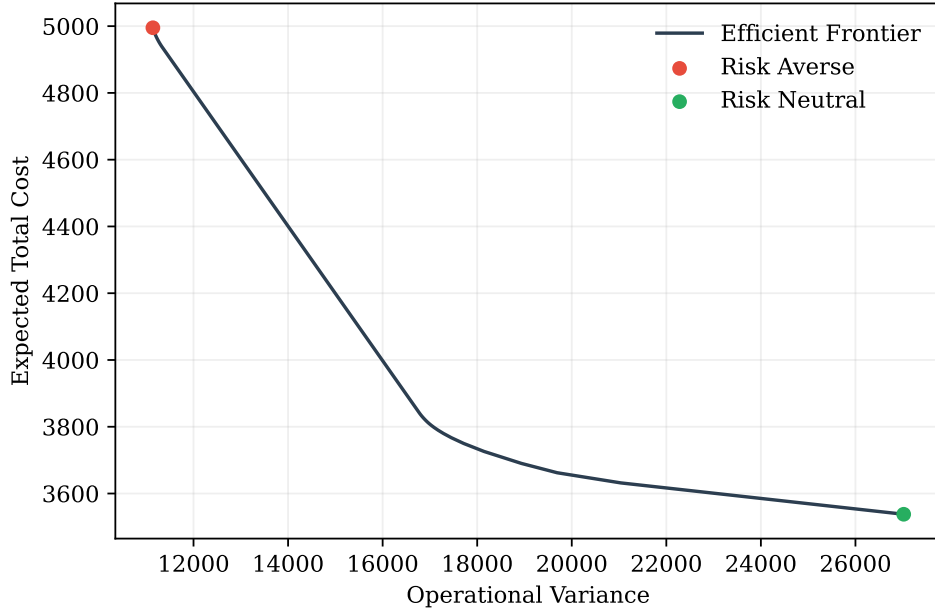


Figure 3.4: The Efficient Frontier for the warehouse location problem. The curve illustrates the trade-off between minimizing expected operational costs and reducing cost variance (risk). The green marker represents a risk-neutral approach ($\lambda = 0$), while the red marker indicates increasing risk aversion.

3.3 Distributionally Robust Warehouse Location Optimization

Readers of the first sections should be feeling slightly uneasy regarding the sensitivity of these optimization problems. By sensitivity, we mean how the optimal solution changes when the input data changes. For example, there is a lot of expected values for the demand $E(D_j)$, that as we have seen, when there is a small number of observations or it is in reality time-varying, the sample mean $\sum_{t=1}^T d_t$ might be within an range of the true value. Compounding this effect across all expected values and variances (which are in turn expected values as well) creates an effect where the optimal solution of solving the warehouse problem with a sample observations $\hat{q}_{ij}^*$ might be substantially different from the true optimal solution q_{ij}^* .

Given a fixed solution q_{ij}^* we have explored previously how to build confidence intervals using the CLT (see Chapter 2 and the empirical distributions section to see some of these techniques). To get a feel for the interplay between the sample size n and the number of vertices N we use a different but useful way of building confidence intervals, we illustrate this with an example:

Example 3.1 (Local fulfillment centers). A last-mile delivery company wants to know the

optimal location to build their fulfillment centers to serve the New York metropolitan area. Geographically, this area is composed by 23 counties, which makes a graph with $N = 23$ nodes. The company has a sample size of daily orders for each of the counties going over 4 years for a total of around $n = 1,460$ observations. Their data-scientist solves the basic warehouse location problem in Equation (3.5) obtaining a solution x_i^*, q_{ij}^* for the optimal locations of the warehouses and the quantity that each should deliver to each county. The management of the company seems skeptical of the solution and ask the data-scientist for confidence intervals for their estimation. Specifically, they want to know with $1 - \alpha = 95\%$ confidence what is a range for the variable cost component of the cost (involving the quantities q_{ij}), as they know with certainty how much it costs to run each of the facilities once active.

To solve this problem, we use a very useful result known as *Hoeffding's Inequality* which states the following for a sum of independent random variables $S_n = X_1 + \dots + X_n$, where all variables are bounded, that is, $P(X_i \in [\ell, u]) = 1$. Then, we have⁵

$$P(S_n - E[S_n] \geq \varepsilon) \leq \exp\left(\frac{-2\varepsilon^2}{n(u - \ell)^2}\right), \quad (3.8)$$

The trick to use this bound is to express the objective value as a function operating on a real valued function. We initially have a random vector for the demands $\mathbf{D} = (D_1, \dots, D_N)$, given a solution $x = (x_1, \dots, x_M)$ and $\mathbf{q} = (q_{11}, \dots, q_{MN})$ with cost vector $c = (c_{11}, \dots, c_{MN})$. The value that we want to bound is $E[c\mathbf{q}^\top]$. The first observation is noting that for any solution, in practice it should be the case that $\sum_{i \in W} q_{ij} = D_j$ for all nodes $j \in V$. Then, by letting the vector $\bar{c} = (\bar{c}_1, \dots, \bar{c}_N)$ where $\bar{c}_j = (\sum_{i \in W} c_{ij} q_{ij}) / (\sum_{i \in W} q_{ij})$ for $j \in V$. In other words, the vector $\bar{c}_j$ is the average cost to deliver to each location. With this, we have $c\mathbf{q}^\top = \bar{c}\mathbf{D}^\top$. Now we can use Hoeffding's inequality to build our cost bound. The function $\bar{c}\mathbf{D}^\top$ can be seen as the generic random variable X in the inequality. The first step is to determine its bound. Note that all demands are bounded below by 0, that is, $\ell = 0$. Likewise, each demand is bounded by above by $\bar{d}_j$, then, all of them are bounded by $\bar{d} = \max_{j \in V} \bar{d}_j$. Then, it follows that the cost is bounded as $\bar{c}\mathbf{D}^\top \leq |\bar{c}| \bar{d}$ (we abuse notation by letting $|\cdot|$ be the $L - 1$ norm for vectors. For example, $|\bar{c}| = \sum_{j \in V} |\bar{c}_j|$). This norm $|\bar{c}|$ is bounded by $N \max\{c_{ij}\}$ where $\max\{c_{ij}\}$ is the highest possible cost, the N is simply the number of vertices in the graph. Then, the random variable $P(\bar{c}\mathbf{D}^\top \in [0, N \max\{c_{ij}\} \bar{d}]) = 1$.

An i.i.d sample of the demand vector $\mathbf{D}$ is represented as vectors $\mathbf{d}_k$ for $k = 1, \dots, n$, where $\mathbf{d}_k$ is a vector with N elements, the demand for each region. The sample expected cost is

⁵The more general version, states that when each random variable is bounded as $P(X_i \in [\ell_i, u_i]) = 1$ the denominator of the exponential is $\sum_{i=1}^n (u_i - \ell_i)^2$ instead of $n(u - \ell)^2$.

then $\mathbb{E}_{\mathbf{P}_n}[\bar{c}\mathbf{D}] = \frac{1}{n} \sum_{k=1}^n \bar{c}\mathbf{d}_k$ while the true cost is the expected value $\mathbb{E}[\bar{c}\mathbf{D}]$. Using Hoeffding's inequality we get that:

$$\mathbb{P}(\mathbb{E}_{\mathbf{P}_n}[\bar{c}\mathbf{D}] - \mathbb{E}[\bar{c}\mathbf{D}] \geq \epsilon) \leq \exp\left(\frac{-2n^2\epsilon^2}{n(N \max\{c_{ij}\}\bar{d})^2}\right), \quad (3.9)$$

With $1 - \alpha = 95\%$ probability, we get that the true cost is bounded by (solving for ϵ):

$$\mathbb{E}[\bar{c}\mathbf{D}] \leq \mathbb{E}_{\mathbf{P}_n}[\bar{c}\mathbf{D}] + N \max\{c_{ij}\}\bar{d} \sqrt{\frac{1}{2n} \ln\left(\frac{1}{\alpha}\right)}. \quad (3.10)$$

This means that solving the problem with a sample of n observations is likely underestimating the true expected cost due to the uncertainty caused by the number of geographical locations N . To mitigate this uncertainty requires at least a quadratic increase in the sample size n . In the next section, we present a strategy to deal with solving an alternative optimization problem that deals with the fact that the true distribution is in a neighborhood of the true distribution.

3.3.1 A reformulation using duality and optimal transport

Back in the first chapter, we defined both the actual expected value of a random variable and their sample counterpart. A specially important random variable is the indicator function $\mathbf{1}_e$ that is equal to 1 if event e occurs, and 0 otherwise. e could be for example the event that today rains, or that the demand exceeds a thousand units. The expected value of an indicator $\mathbb{E}(\mathbf{1}_e)$ is equal to the probability of the event $\mathbb{P}(e)$. And the sample estimate is the same as we defined before, that is, $p_n(e) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{e_i\}$. For example, the sample probability that the demand is equal to 10, is the sample probability $p_n(D = 10) = \sum_{i=1}^n \mathbf{1}\{d_i = 10\}/n$. Sample probabilities behave the same as expected values, that is, they converge to their true value if the distribution is i.i.d.

For a random variable X taking values in the set $\mathcal{X}$ (for example, in the case of the demand random variable D , the demand takes values in the set of the natural numbers $\mathbb{N}$), the collection of sample probabilities $p_n(X = x) = \sum_{i=1}^n \mathbf{1}\{X_i = x\}$ for all $x \in \mathcal{X}$, forms an *empirical distribution* of X . This distribution is typically denoted by $\mathbf{P}_n$ instead of the true probability distribution $\mathbf{P}$. The reader can already imagine that as the sample size n increases, the empirical distribution will converge to the true one by the LLN, that is $\mathbf{P}_n \rightarrow \mathbf{P}$. It is also, expected that the empirical distribution of n observations will overestimate the probability of some events, while also underestimating others. In fact, since both probability distributions add up to 1, the amount of overestimation is exactly the amount of underestimation (think of why this is the case).

Given these facts, the methodology known as *Distributionally Robust Optimization* (DRO) aims to incorporate the idea that the true distribution $\mathbf{P}$ of the data is within some neighborhood of the empirical distribution $\mathbf{P}_n$. Here, neighborhood is taken as a *distance metric* between

the true and empirical distribution. Let $\mathcal{D}$ denote a metric distance function between distributions⁶. Then, we want to consider distributions $\mathbb{P}$ that are around the observed empirical distribution $\mathbb{P}_n$ within a distance δ , that is, distributions $\mathbb{P} : \mathcal{D}(\mathbb{P}_n, \mathbb{P}) \leq \delta$. In plain terms, a decision-maker acknowledges that their finite data samples might bias their estimates and wants to optimize considering the case where the true distribution is in some neighborhood of the empirical distribution $\mathbb{P}_n$, in this case, neighborhood means the set of possible distributions that are in a *ball* of radius δ around $\mathbb{P}_n$. The decision-maker picks the radius according to their risk aversion (similar to choosing λ in the previous examples). A larger radius δ means less trust in the empirical distribution (for example, due to a low sample size) while a small radius means that there is high confidence in the empirical distribution. See Figure 3.5 for a conceptual representation of the DRO approach.

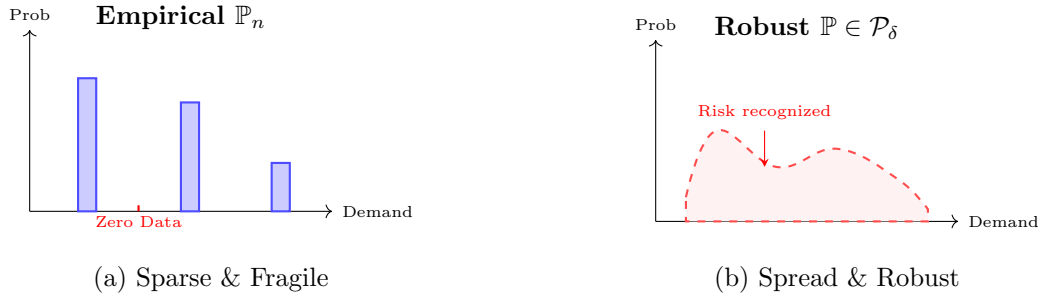


Figure 3.5: Comparison of fulfillment modeling. The empirical approach (left) ignores events not present in the sample, leading to $q_{ij} = 0$ for potentially critical regions. The DRO approach (right) considers a distribution that spreads probability mass into those gaps, forcing the optimization to allocate resources ($q_{ij} > 0$) for unobserved but plausible demand.

The DRO formulation can be set up in many ways. We will focus in a paradigm that uses duality theory and optimal transport. To simplify the problem we will only consider variation in the demand, that is, the costs c_{ij} are assumed to be known and fixed. Moreover, the historical (sample) demand is the matrix $D_n = (\mathbf{d}_k)$ for $k = 1, \dots, n$, where $\mathbf{d}_k$ is a vector with the historical demands for each $i \in V$, that is, $\mathbf{d}_k$ is a vector of size N . Each of these vectors is an i.i.d. sample of the vector $\mathbf{D}$. Denote the set of distributions within δ distance of the empirical distributions as $\mathcal{P}_\delta = \{\mathbb{P} : \mathcal{D}(\mathbb{P}_n, \mathbb{P}) \leq \delta\}$. Let $x = (x_1, \dots, x_M)$. First, we re-write the warehouse problem in this setting as:

$$\min \left\{ \tilde{c}x^\top + \sup_{\mathcal{P}_\delta} \mathbb{E} [\min c\mathbf{q}^\top | A\mathbf{q}^\top + Bx^\top \geq (\mathbf{D}, \mathbf{0})^\top] \right\}, \quad (3.11)$$

This formulation is a matrix form re-write of the problem of interest that is going to be useful for some algebraic manipulations that we are performing. For example, $\tilde{c}x^\top$ is equal to $\sum_{i \in W} c_i x_i$

⁶A distance metric is a function $\mathcal{D} : \mathcal{M} \times \mathcal{M} \rightarrow \mathbb{R}$ that operates on a set $\mathcal{X}$ with the properties: For $p_1, p_2, p_3 \in \mathcal{M}$ it is satisfied $\mathcal{D}(p_1, p_1) = 0$, $\mathcal{D}(p_1, p_2) > 0$ for $p_1 \neq p_2$, $\mathcal{D}(p_1, p_2) = \mathcal{D}(p_2, p_1)$ for all $p_1, p_2 \in \mathcal{M}$ and $\mathcal{D}(p_1, p_3) \leq \mathcal{D}(p_1, p_2) + \mathcal{D}(p_2, p_3)$ which is the so-called triangle inequality.

and $c\mathbf{q}^\top = \sum_{i \in W} \sum_{j \in V} c_{ij} q_{ij}$. The constraints $A\mathbf{q}^\top + Bx^\top \geq (\mathbf{D}, \mathbf{0})^\top$ are the usual constraints of the problem, such as demand satisfaction, e.g. see the constraints in (3.5), A would be the matrix of coefficients multiplying the $\mathbf{q}$ variables, while the right hand side is the random demand $\mathbf{D}$ for the demand satisfaction constraints and $\mathbf{0}$ for the constraints that only depend on variables. The interpretation of the problem is quite similar: minimize the overall cost of building the facilities, subject to the worst distribution $\mathcal{P}_\delta$. Typically, the intuitive explanation is that nature chooses the worst distribution given x (the sup part) and then the optimization chooses the optimal quantities $\mathbf{q}$.

The first step to solve this problem is to re-formulate the sup by exploiting duality and results from Transport Theory. Letting $Q(x, \mathbf{D}) = \min_{\mathbf{q} \geq 0} c\mathbf{q}^\top | A\mathbf{q}^\top + Bx^\top \geq (\mathbf{D}, \mathbf{0})^\top$, we are going to write the Lagrangian of $\sup_{\mathcal{P}_\delta} \mathbb{E}[Q(x, \mathbf{D})]$ (noting that the definition of $\mathcal{P}_\delta$ is a constraint defined by the distance of the distributions), for a pair of variables $\mathbf{P}, \lambda$:

$$\sup_{\mathcal{P}_\delta} \mathbb{E}[Q(x, \mathbf{D})] = \min_{\lambda \geq 0} \sup_{\mathbf{P}} \{ \mathbb{E}_{\mathbf{P}}[Q(x, \mathbf{D})] + \lambda(\delta - \mathcal{D}(\mathbf{P}, \mathbf{P}_n)) \}, \quad (3.12)$$

The next step is to choose a distance metric $\mathcal{D}$ that makes the problem tractable. One such metric, is the so-called *Wasserstein-1* distance. Choosing this distance $\mathcal{D} = W_1$ has the following property for any function $g(\mathbf{D})$:

$$\sup_{\mathbf{P}} [\mathbb{E}_{\mathbf{P}}[g(\mathbf{D})] - \lambda W_1(\mathbf{P}, \mathbf{P}_n)] = \mathbb{E}_{\mathbf{P}_n} \left[\sup_{\mathbf{D}} \{ g(\mathbf{D}) - \lambda |\mathbf{D} - \mathbf{D}_k| \} \right], \quad (3.13)$$

Where $\mathbf{D}_k$ is a generic sample of the empirical distribution $\mathbf{P}_n$ and $|\cdot|$ for vectors denotes the $L - 1$ norm. We can let $g(\mathbf{D}) = Q(x, \mathbf{D})$. The math behind the result is somewhat technical and outside the scope of this section. By letting $g(\mathbf{D}) = Q(x, \mathbf{D})$ and rearranging $\min_{\lambda \geq 0} \sup_{\mathbf{P}} \{ \mathbb{E}_{\mathbf{P}}[Q(x, \mathbf{D})] + \lambda(\delta - \mathcal{D}(\mathbf{P}, \mathbf{P}_n)) \}$ as $\min_{\lambda \geq 0} (\lambda\delta + \sup_{\mathbf{P}} \{ \mathbb{E}_{\mathbf{P}}[Q(x, \mathbf{D})] - \lambda\mathcal{D}(\mathbf{P}, \mathbf{P}_n) \})$ combining the two equations yields:

$$\sup_{\mathcal{P}_\delta} \mathbb{E}[Q(x, \mathbf{D})] = \min_{\lambda \geq 0} \left\{ \lambda\delta + \mathbb{E}_{\mathbf{P}_n} \left[\sup_{\mathbf{D}} \{ Q(x, \mathbf{D}) - \lambda |\mathbf{D} - \mathbf{D}_k| \} \right] \right\}, \quad (3.14)$$

Which is great news as we know how to estimate expected values in the empirical distribution, combining the original objective we get:

$$\min_{\lambda \geq 0} \left\{ \tilde{c}x^\top + \lambda\delta + \frac{1}{n} \sum_{k=1}^n \sup_{\mathbf{D}} \{ Q(x, \mathbf{D}) - \lambda |\mathbf{D} - \mathbf{d}_k| \} \right\}, \quad (3.15)$$

Recalling $Q(x, \mathbf{D}) = \min_{\mathbf{q} \geq 0} c\mathbf{q}^\top | A\mathbf{q}^\top + Bx^\top \geq (\mathbf{D}, \mathbf{0})^\top$, we can calculate the dual of this problem as $\max_{\boldsymbol{\eta} \geq 0} \boldsymbol{\eta}[(\mathbf{D}, \mathbf{0})^\top - Bx^\top] | \boldsymbol{\eta}A \leq c$. Each of the terms $\sup_{\mathbf{D}} \{ Q(x, \mathbf{D}) - \lambda |\mathbf{D} - \mathbf{d}_k| \}$ can be then written as $\sup_{\mathbf{D}} \{ \boldsymbol{\eta}[(\mathbf{D}, \mathbf{0})^\top - Bx^\top] - \lambda |\mathbf{D} - \mathbf{d}_k| \}$. Exchanging the sup and the max and isolating the terms depending on the vector $\mathbf{D}$ yields:

$$\max_{\boldsymbol{\eta} \geq 0, \boldsymbol{\eta}A \leq c} -\boldsymbol{\eta}Bx^\top + \sup_{\mathbf{D}} \{ \boldsymbol{\eta}_N \mathbf{D}^\top - \lambda |\mathbf{D} - \mathbf{d}_k| \}, \quad (3.16)$$

Where $\boldsymbol{\eta}_N$ are the first N elements of $\boldsymbol{\eta}$ corresponding to the dual variables of the demand satisfaction constraints of the warehouse problem. Next, define the variable $\mathbf{v}_k = \mathbf{D} - \mathbf{d}_k$. The

inner problem $\sup_{\mathbf{D}} \{\boldsymbol{\eta}_N \mathbf{D}^\top - \lambda \|\mathbf{D} - \mathbf{d}_k\|\}$ can be re-written as $\boldsymbol{\eta}_N \mathbf{d}_k^\top + \sup_{\mathbf{v}_k} \{\boldsymbol{\eta}_N \mathbf{v}_k^\top - \lambda \|\mathbf{v}_k\|\}$. The last term is the convex conjugate of the $L - 1$ norm and is equal to 0 when $|\max \boldsymbol{\eta}_N| \leq \lambda$ or equal to ∞ otherwise. That it is equal to 0 is achieved by the constraints $-\mathbf{1}\lambda \leq \boldsymbol{\eta}_N \leq \mathbf{1}\lambda$. See Figure 3.6 for the intuition behind these constraints.

With this we can finally finish the rewrite of the problem as:

$$\begin{aligned} \min_{\lambda, \boldsymbol{\eta}, x} \quad & \left\{ \tilde{c}x^\top + \lambda\delta + \frac{1}{n} \sum_{k=1}^n (\boldsymbol{\eta}_N \mathbf{d}_k^\top - \boldsymbol{\eta} B x^\top) \right\} \\ \text{s.t.} \quad & \boldsymbol{\eta} A \leq c, -\mathbf{1}\lambda \leq \boldsymbol{\eta}_N \leq \mathbf{1}\lambda, \\ & \lambda \geq 0, \boldsymbol{\eta} \geq 0, x \in \{0, 1\}^M. \end{aligned} \quad (3.17)$$

Which is almost a MIP except for the variables $\boldsymbol{\eta} B x^\top$. The standard way of dealing with this, is to define new variables $z_{ij} = \eta_i x_j$ and add the constraints $z_{ij} \leq \tilde{M} x_j, z_{ij} \leq \eta_i, z_{ij} \geq \eta_i - \tilde{M}(1 - x_j), z_{ij} \geq 0$ for a large enough $\tilde{M}$ (this is commonly known as the big-M method).

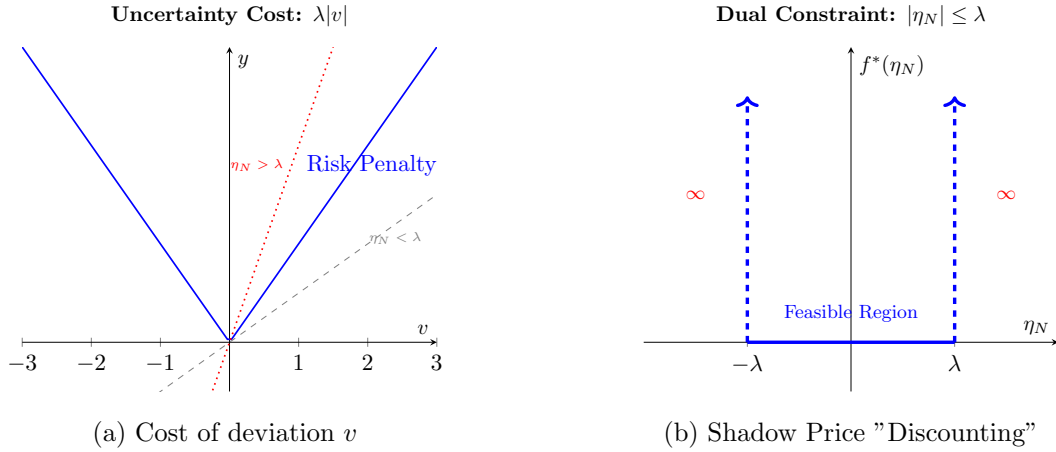


Figure 3.6: The relationship between deviation cost and shadow prices. In (a), if the shadow price η_N (the marginal cost of demand) is steeper than the risk penalty λ , the sup-problem becomes unbounded. In (b), this translates to a hard constraint: the robust optimizer only considers shadow prices "discounted" within the $[-\lambda, \lambda]$ interval, effectively capping the system's sensitivity to demand uncertainty.